

Computational Simulation of the Wilberforce Pendulum: normal modes, energy conservation and a strategy for measuring the coupling constant

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Abstract. The Wilberforce pendulum is a mechanical oscillator in which a helical spring couples the longitudinal motion (vertical translation) and torsional motion (axial rotation) of a suspended mass. Under resonance, when the two natural frequencies coincide, the system periodically exchanges all its energy between the two modes, producing the characteristic *beating*. In this work we solve the system's coupled differential equations with the fourth-order Runge–Kutta method implemented in a modular Fortran 2008 package using quadruple precision, and compare the results with the analytical normal-mode solution, an independent adaptive integrator and data from the literature. The solver's relative error against the closed-form solution is $\sim 10^{-6}$, and total energy is conserved with a relative drift of $\sim 5.5 \times 10^{-8}$ over 4000 steps; the integrator's empirical convergence order is 4.00. As a contribution, we propose and numerically validate a strategy for measuring the coupling constant ε — which belongs not to the spring but to the motion — from a single period measurement, recovering it with an error below 0.1%. The project also includes an interactive three-dimensional simulator of the pendulum.

Keywords: Coupled oscillations • Wilberforce pendulum • Normal modes • Runge–Kutta • Numerical simulation • Fortran • Coupling constant.

1 Introduction

Oscillatory systems occur throughout nature: periodic chemical reactions, electronic circuits, biological and biochemical rhythms, and astronomical bodies such as asteroids, planets and stars. When two or more oscillators are coupled, collective phenomena — synchronization, energy transfer and beating — emerge, characterized by phase differences and the ratio of their periods.

Among coupled oscillators, the *Wilberforce pendulum* has particular educational value because it visibly demonstrates the pairing of torsional and longitudinal oscillations in a mass–spring system and illustrates the conservation of mechanical energy. It consists of a helical spring fixed at its upper end, from which a cylindrical mass is suspended. This mass can both translate vertically along the spring's axis and rotate about it (Figure 1). Because the winding of the helix links extension to torsion, the two degrees of freedom exchange energy.

This article has two objectives. First, to develop a rigorous numerical description of the system's dynamics by solving its coupled differential equations with fourth-order Runge–Kutta and to validate it extensively against the analytical solution of the resonant case, an independent numerical method and the lit-

erature. Second, to analyze the system's behavior as the coupling constant ε varies and propose a physical strategy for measuring it. The code, written as a modular scientific package, together with the analysis layer, interactive three-dimensional simulator and this document, forms a project reproducible from end to end.

2 Physical model

2.1 Lagrangian and equations of motion

Neglecting the spring's mass, the kinetic energy is the sum of the linear and rotational contributions, $T = \frac{1}{2}m\dot{z}^2 + \frac{1}{2}I\dot{\theta}^2$, while the potential energy combines the longitudinal elastic energy, torsional energy and a bilinear coupling term that models the helix:

$$V = \frac{1}{2}kz^2 + \frac{1}{2}\delta\theta^2 + \frac{1}{2}\varepsilon z\theta. \quad (1)$$

The Lagrangian $L = T - V$ is

$$L = \frac{1}{2}m\dot{z}^2 + \frac{1}{2}I\dot{\theta}^2 - \frac{1}{2}kz^2 - \frac{1}{2}\delta\theta^2 - \frac{1}{2}\varepsilon z\theta, \quad (2)$$

where ε is the coupling constant. Applying the Euler–Lagrange equations $\frac{d}{dt}(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$ to $q \in$

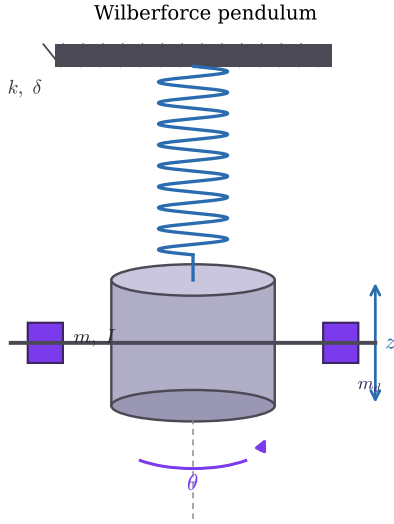


Figure 1: Diagram of the Wilberforce pendulum. A cylindrical body of mass m and moment of inertia I hangs from a helical spring with longitudinal constant k and torsional constant δ . The rod with sliding masses m_d allows I to be adjusted experimentally. The degrees of freedom are vertical translation z and axial rotation θ .

$\{z, \theta\}$ (see Appendix A) yields the coupled system

$$m \ddot{z} + k z + \frac{1}{2} \varepsilon \theta = 0, \quad (3)$$

$$I \ddot{\theta} + \delta \theta + \frac{1}{2} \varepsilon z = 0. \quad (4)$$

Solving for the accelerations gives the form integrated by the solver:

$$\ddot{z} = -\frac{k}{m}z - \frac{\varepsilon}{2m}\theta, \quad \ddot{\theta} = -\frac{\delta}{I}\theta - \frac{\varepsilon}{2I}z. \quad (5)$$

2.2 Resonance

The uncoupled oscillators would have natural frequencies $\omega_z^2 = k/m$ and $\omega_\theta^2 = \delta/I$. The Wilberforce phenomenon is fully manifested at *resonance*, that is, when the two coincide, $\omega_z = \omega_\theta \equiv \omega_0$. This imposes

$$\frac{k}{m} = \frac{\delta}{I} \implies k = \frac{m\delta}{I}, \quad \omega_0^2 = \frac{\delta}{I}. \quad (6)$$

The package enforces (6) exactly when loading each configuration, so the only free frequency of the uncoupled problem is ω_0 .

2.3 Normal modes and beating

Seeking solutions of the form $e^{i\omega t}$ in (3)–(4) and diagonalizing at resonance, the system has two normal-mode frequencies symmetrically split about ω_0 :

$$\omega_1^2 = \omega_0^2 + \frac{\varepsilon}{2\sqrt{mI}}, \quad \omega_2^2 = \omega_0^2 - \frac{\varepsilon}{2\sqrt{mI}}. \quad (7)$$

The superposition of two nearby frequencies produces *beating*: energy oscillates between the modes at frequency $\omega_{\text{beat}} = \omega_1 - \omega_2$ and period

$$T_{\text{beat}} = \frac{2\pi}{\omega_1 - \omega_2} \approx \frac{4\pi\omega_0\sqrt{mI}}{\varepsilon} \quad (\varepsilon \text{ small}). \quad (8)$$

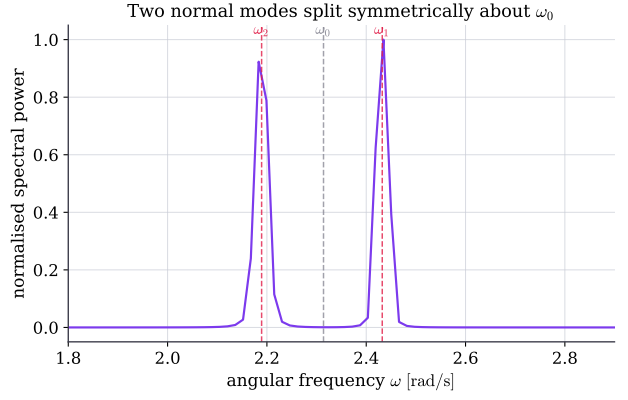


Figure 2: Spectrum of $\theta(t)$ for the reference system. The two normal modes appear as peaks symmetrically split about the natural frequency ω_0 , as predicted by equation (7).

Figure 2 numerically confirms the splitting in (7): the Fourier transform of $\theta(t)$ displays two clear peaks at ω_1 and ω_2 , symmetric about ω_0 .

2.4 Analytical solution

For the initial conditions used throughout this work — release from rest, $\theta(0) = \theta_0$, $\dot{\theta}(0) = 0$, $z(0) = z_0$, $\dot{z}(0) = 0$ — the closed-form solution of the resonant case is

$$\theta(t) = B \cos \omega_1 t + D \cos \omega_2 t, \quad (9)$$

$$z(t) = \frac{2I}{\varepsilon} [B(\omega_1^2 - \omega_0^2) \cos \omega_1 t + D(\omega_2^2 - \omega_0^2) \cos \omega_2 t], \quad (10)$$

with coefficients fixed by the initial conditions,

$$B = \frac{\theta_0(\omega_2^2 - \omega_0^2) - z_0 \varepsilon / 2I}{\omega_2^2 - \omega_1^2}, \quad D = \frac{z_0 \varepsilon / 2I - \theta_0(\omega_1^2 - \omega_0^2)}{\omega_2^2 - \omega_1^2}. \quad (11)$$

The complete derivation is given in Appendix A.

2.5 Energy balance

The instantaneous mechanical energy is decomposed into five contributions,

$$E = \underbrace{\frac{1}{2}m\dot{z}^2 + \frac{1}{2}kz^2}_{E_z} + \underbrace{\frac{1}{2}I\dot{\theta}^2 + \frac{1}{2}\delta\theta^2}_{E_\theta} + \underbrace{\frac{1}{2}\varepsilon z\theta}_{E_c}, \quad (12)$$

where E_z , E_θ and E_c are the longitudinal, torsional and coupling energies. Conservation of E is the principal indicator of the quality of the numerical integration.

3 Numerical method and implementation

3.1 Fourth-order Runge–Kutta

We write the system (5) as a first-order problem for the state vector $\mathbf{u} = (z, \dot{z}, \theta, \dot{\theta})$, $\dot{\mathbf{u}} = \mathbf{f}(\mathbf{u})$, and advance it

with the classical fourth-order Runge–Kutta scheme at fixed step h :

$$\begin{aligned} k_1 &= f(u_n), & k_2 &= f(u_n + \frac{h}{2}k_1), \\ k_3 &= f(u_n + \frac{h}{2}k_2), & k_4 &= f(u_n + h k_3), \\ u_{n+1} &= u_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned} \quad (13)$$

The local error per step is $\mathcal{O}(h^5)$ and the global error is $\mathcal{O}(h^4)$. The state is integrated in *quadruple precision* (`real*16`, approximately ~ 33 significant digits) so that rounding remains far below the truncation error. Listing 1 shows the integration step; note how it reproduces (13) term by term for both coordinates simultaneously.

Listing 1: Fourth-order Runge–Kutta step for the coupled system (`src/wilberforce_integrator.f90`).

```
1 pure function rk4_step(p, u, h) result(un)
2   type(params_t), intent(in) :: p
3   type(state_t), intent(in) :: u
4   real(qp), intent(in) :: h
5   type(state_t) :: un
6   real(qp) :: z, v, x, b
7   real(qp) :: k1y, k1v, k2y, k2v, k3y, k3v, k4y, k4v
8   real(qp) :: j1x, j1b, j2x, j2b, j3x, j3b, j4x, j4b
9   z=u%z; v=u%v; x=u%theta; b=u%omega
10  ! Etapa 1
11  k1y = deriv_z(v); k1v = accel_z(z, x, p)
12  j1x = deriv_theta(b); j1b = accel_theta(z, x, p)
13  ! Etapa 2 (medio paso)
14  k2y = deriv_z(v + h*k1v/2.)
15  k2v = accel_z(z + h*k1y/2., x + h*j1x/2., p)
16  j2x = deriv_theta(b + h*j1b/2.)
17  j2b = accel_theta(z + h*k1y/2., x + h*j1x/2., p)
18  ! ... etapas 3 y 4 analogas ...
19  un%z = z + (1./6.)*h*(k1y+2.*k2y+2.*k3y+k4y)
20  un%theta = x + (1./6.)*h*(j1x+2.*j2x+2.*j3x+j4x)
21 end function rk4_step
```

3.2 Package architecture

Rather than a monolithic script, the solver is organized as a modular scientific package in Fortran 2008: each responsibility — types and precision, equations of motion, analytical solution, integrator, energy diagnostics, configurations, and input/output — resides in its own module (Table 1). Listing 2 illustrates the equations-of-motion module, a direct transcription of (5). The package is built with `make`, `CMake` or the `fpm` package manager, and includes a suite of regression and physics tests.

Listing 2: Right-hand side of the equations of motion (`src/wilberforce_model.f90`).

```
1 pure function accel_z(z, theta, p) result(a)
2   real(qp), intent(in) :: z, theta
3   type(params_t), intent(in) :: p
4   real(qp) :: a, k, m, e
5   k = real(p%k, qp); m = real(p%m, qp); e = real(p%eps, qp)
6   a = -k/m * z - 0.5_qp*e/m * theta ! z'' = -(k/m)z - (e/2m)theta
7 end function accel_z
```

Table 1: Numerical-core modules and their responsibilities.

Module	Responsibility
<code>kinds</code>	working precisions (<code>dp</code> , <code>qp</code>)
<code>types</code>	derived types <code>params_t</code> , <code>state_t</code>
<code>model</code>	right-hand side, eq. (5)
<code>analytic</code>	normal-mode solution
<code>integrator</code>	RK4 step and simulation driver
<code>energy</code>	energy-balance diagnostics
<code>presets</code>	14 predefined configurations
<code>io</code>	data output, directory creation

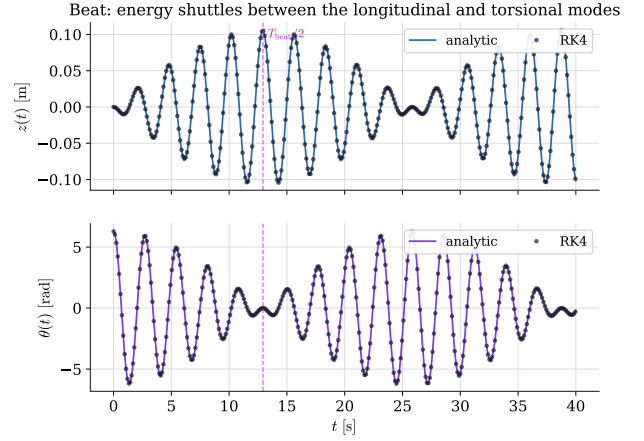


Figure 3: Beating in the reference system. Energy transfers from torsion (bottom) to translation (top) and back with period T_{beat} . The markers (RK4) coincide with the analytical solution (line).

4 Validation

The correctness of the solver is established at four levels: comparison with the analytical solution, a convergence study, conservation of energy, and comparison with independent methods and data.

4.1 Analytical versus numerical solution

Figure 3 shows the evolution of $z(t)$ and $\theta(t)$ for the Berg–Marshall/Hill system ($\delta = 7.4429 \times 10^{-4}$, $\varepsilon = 9.27 \times 10^{-3}$, $m = 0.4905$, $I = 1.39 \times 10^{-4}$; $z_0 = 0$, $\theta_0 = 2\pi$). The system starts in pure torsion and transfers all its energy to translation at $T_{\text{beat}}/2 \approx 12.9$ s, then returns: this is beating. The RK4 integrator’s points overlay the analytical solution without a visible deviation.

4.2 Fourth-order convergence

Integrating with decreasing steps h and measuring the maximum error against the analytical solution gives Figure 4. The error scales as h^4 over seven decades; the fitted empirical order is 4.00, in exact agreement with the theory of the method.

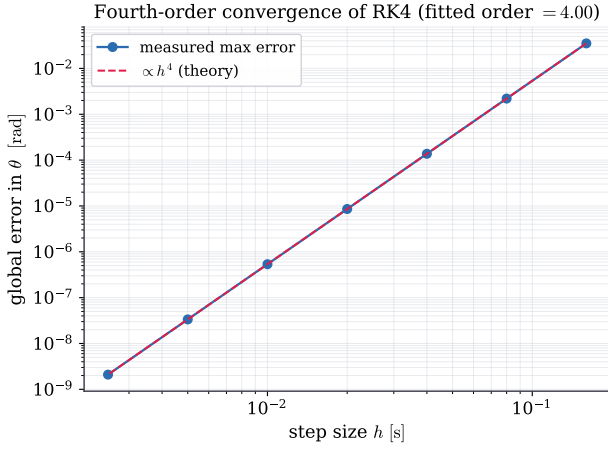


Figure 4: Integrator convergence: the global error in θ decreases as h^4 , confirming the fourth order of the Runge-Kutta scheme.

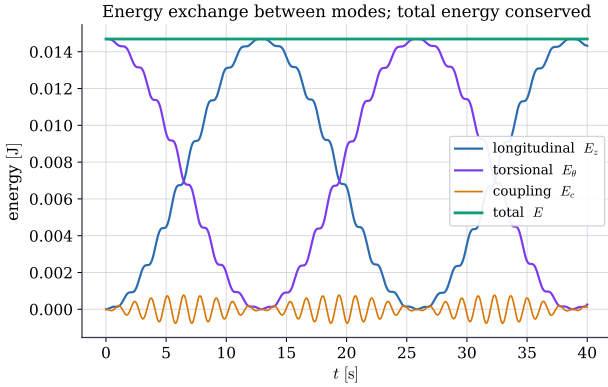


Figure 5: Energy exchange between modes. E_z and E_θ oscillate in antiphase; E_c remains small and total energy is conserved.

4.3 Conservation of energy

Figure 5 shows the contributions E_z , E_θ , E_c and total energy. The longitudinal and torsional energies exchange their content in antiphase while the total remains flat. The absolute drift remains bounded at $\sim 8 \times 10^{-10}$ J ($\sim 5.5 \times 10^{-8}$ relative) over 4000 steps, constituting the main evidence of the fidelity of the integration.

4.4 Cross-validation and literature

To rule out errors in both the solver and the analytical solution, we compare four descriptions of the same system (Table 2 and Figure 6): the Fortran core (RK4, quadruple precision), an independent Python reimplementation (RK4, double precision), a high-order adaptive integrator (DOP853) and the closed-form solution. The independent method reproduces the analytical solution to $\sim 10^{-11}$, confirming its correctness; the Fortran solver reproduces it with relative error $\sim 10^{-6}$ ($\Delta\theta \approx 10^{-5}$ absolute), a level set by the step $h = 0.01$ and the single-precision weight literals inherited from the original code. The parameters and initial condi-

Table 2: Cross-validation for the reference system (amplitudes $\theta \sim 6.28$ rad, $z \sim 0.106$ m).

Comparison	Δz	$\Delta\theta$
Fortran RK4 vs. Python RK4	2.9×10^{-7}	1.2×10^{-5}
Fortran RK4 vs. DOP853	2.6×10^{-7}	1.1×10^{-5}
Fortran RK4 vs. analytical	2.6×10^{-7}	1.1×10^{-5}
DOP853 vs. analytical	4.3×10^{-13}	1.8×10^{-11}

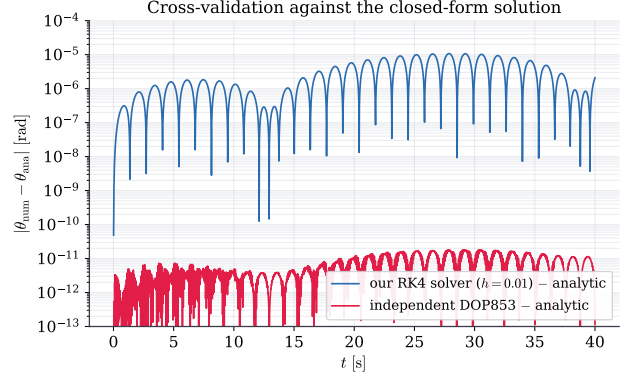


Figure 6: Residuals against the analytical solution. The solver (blue line) agrees to $\sim 10^{-5}$ in absolute value ($\sim 10^{-6}$ relative); the independent DOP853 method (red), to $\sim 10^{-11}$, certifying the closed-form solution.

tions coincide with those in the work of C. Hill [5] and with the measurements of Berg and Marshall [1], yielding identical results.

5 Coupling phenomenology

The configuration plane θ - z (Figure 7) traces the signature of the Wilberforce pendulum: a family of nested ellipses rotating from horizontal (energy in θ) to vertical (energy in z) as the beat progresses. The coupling constant ε governs the speed of this exchange.

Varying ε traverses a continuum of regimes (Figure 8):

- **Strong coupling** (large ε): the modes mix intensely and energy transfer occurs over short times.
- **Weak coupling** (small ε): the modes behave almost independently; transfer is slow and the oscillations and rotations become more distinguishable.
- **Intermediate regime:** a gradual transition between the two extremes.

The beat period diverges as $T_{\text{beat}} \propto 1/\varepsilon$ when $\varepsilon \rightarrow 0$, in excellent agreement with the prediction in (8).

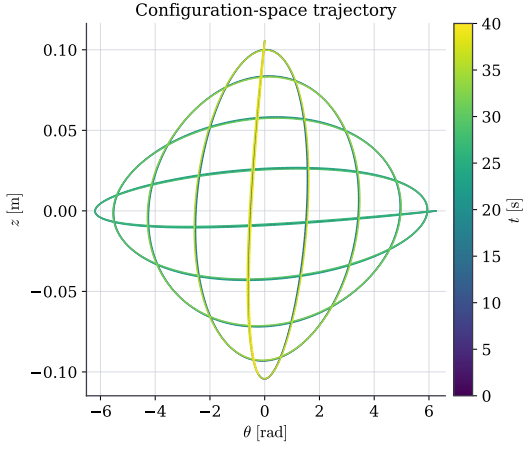


Figure 7: Trajectory in configuration space θ - z , colored by time. The nested ellipses reveal the periodic transfer of energy between the modes.

6 Measuring the coupling constant

The elastic constants k and δ belong to the spring, and the mass m and inertia I are known. By contrast, the coupling constant ε belongs not to the spring but to the coupled *motion*, raising the question: given a real Wilberforce pendulum, how can ε be determined?

The general motion (9) superposes two incompatible frequencies and is therefore not periodic: a period cannot be read off. The idea is to *eliminate one of the modes*. Imposing $D = 0$ in (11), and using $\omega_1^2 - \omega_0^2 = \varepsilon/2\sqrt{mI}$, the condition reduces to a restriction on the initial amplitudes:

$$\theta_0 = z_0 \sqrt{\frac{m}{I}} \quad (14)$$

Prepared in this way, the system oscillates at a single frequency ω_1 with a measurable period $T = 2\pi/\omega_1$. Inverting $\omega_1^2 = \omega_0^2 + \varepsilon/2\sqrt{mI}$:

$$\varepsilon = 2\sqrt{mI} \left(\frac{4\pi^2}{T^2} - \frac{\delta}{I} \right) \quad (15)$$

Figure 9 validates the strategy: starting from $\theta_0 = 0$, the motion contains two frequencies (nonperiodic); applying (14) collapses the spectrum to a single line and makes the motion periodic. Measuring T directly from the trajectory (using interpolated zero crossings) and applying (15), we recover the constant with an error below 0.1% for three different systems. For the system with $\delta = 1.6$, $m = 0.07$, $I = 0.5$, $\varepsilon = 0.075$ (Set 11/12 of the package), with $z_0 = 1$ and $\theta_0 = z_0\sqrt{m/I} = 0.3742$, the measurement $T = 3.4073$ s recovers $\varepsilon = 0.07500$ against the true value 0.07500. Note that this system differs from the reference system (Berg–Marshall) used in the preceding sections.

7 Interactive three-dimensional simulator

As an educational complement, the project includes a self-contained interactive three-dimensional simulator (rendered with WebGL/Three.js embedded in a single file), which integrates the same equations (5) in real time with scheme (13). The user adjusts $\delta, \varepsilon, m, I$ and the initial conditions, observes the helical spring stretching and the cylindrical mass rotating, and follows $z(t)$, $\theta(t)$, the phase plane and energy balance in synchronized panels. Figure 10 shows a sequence generated by the simulator’s headless renderer over a beat: the mass transitions from maximum rotation with zero translation to maximum translation with minimum rotation, making the transfer of energy tangible.

8 Conclusions

We demonstrate that a Lagrangian description of the Wilberforce pendulum, solved numerically with fourth-order Runge–Kutta in quadruple precision, faithfully predicts its motion, velocities and energy without requiring prior analytical development. Validation was extensive: relative agreement with the closed-form solution of $\sim 10^{-6}$, convergence order 4.00, conservation of energy with relative drift $\sim 5.5 \times 10^{-8}$, and agreement with an independent method and the literature. Conservation of energy was our guiding indicator for interpreting the dynamics.

Analysis of the coupling parameter revealed the continuum from strong to weak coupling and the divergence of the beat period as $\varepsilon \rightarrow 0$. As a contribution, we proposed and validated a strategy for measuring the coupling constant — which belongs not to the system but to the motion — from a single period measurement, recovering it with error below 0.1%. Finally, the modular package, reproducible analysis layer and interactive three-dimensional simulator make the project a complete tool for studying this phenomenon.

A Equations of motion

Euler–Lagrange equations. From the Lagrangian (2), for coordinate z ,

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{z}} \right) = \frac{d}{dt} (m\dot{z}) = m\ddot{z}, \quad \frac{\partial L}{\partial z} = -kz - \frac{1}{2}\varepsilon\theta,$$

and similarly for θ , $\frac{d}{dt}(\partial L/\partial \dot{\theta}) = I\ddot{\theta}$ and $\partial L/\partial \theta = -\delta\theta - \frac{1}{2}\varepsilon z$. Substitution into $\frac{d}{dt}(\partial L/\partial \dot{q}) - \partial L/\partial q = 0$ recovers the equations of motion (3)–(4).

Normal-mode frequencies. With the harmonic *ansatz* $z = \hat{z}e^{i\omega t}$, $\theta = \hat{\theta}e^{i\omega t}$, equations (3)–(4) become an algebraic system,

$$\begin{pmatrix} k - m\omega^2 & \varepsilon/2 \\ \varepsilon/2 & \delta - I\omega^2 \end{pmatrix} \begin{pmatrix} \hat{z} \\ \hat{\theta} \end{pmatrix} = \mathbf{0}. \quad (16)$$

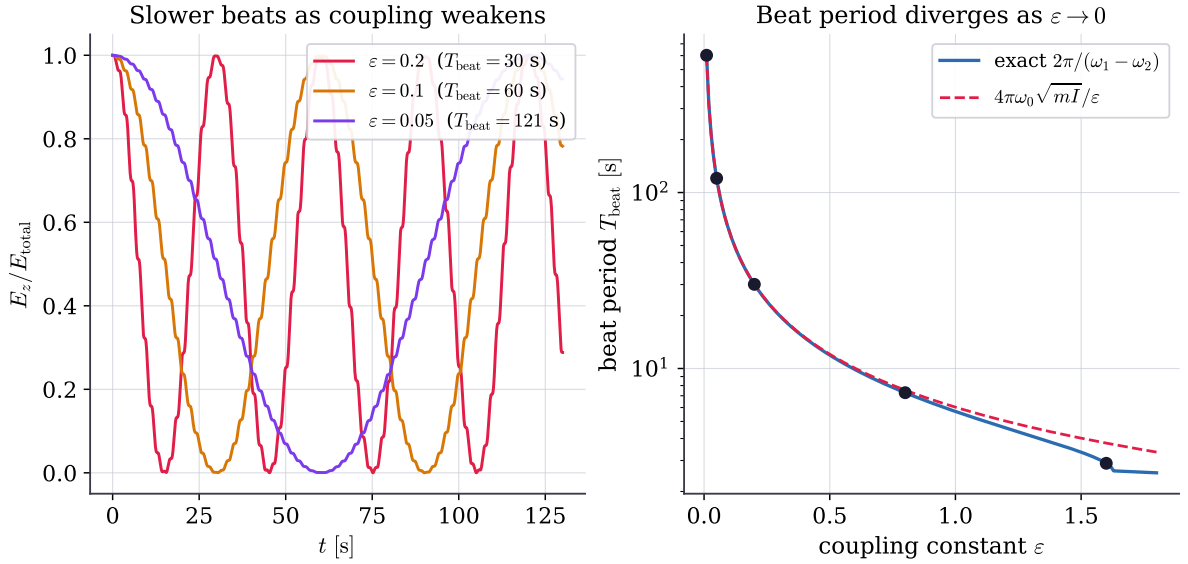


Figure 8: (Left) longitudinal-energy fraction over time: the beat period grows as coupling weakens. (Right) beat period against ε : the exact formula $2\pi/(\omega_1 - \omega_2)$ and the small-coupling approximation $4\pi\omega_0\sqrt{mI}/\varepsilon$ agree except for large ε .

A nontrivial solution requires a zero determinant, $(k - m\omega^2)(\delta - I\omega^2) - \varepsilon^2/4 = 0$. At resonance $k = m\omega_0^2$, $\delta = I\omega_0^2$, the determinant factorizes,

$$mI(\omega_0^2 - \omega^2)^2 - \frac{\varepsilon^2}{4} = 0 \implies \omega^2 = \omega_0^2 \pm \frac{\varepsilon}{2\sqrt{mI}}, \quad (17)$$

yielding the two normal-mode frequencies ω_1, ω_2 in (7).

General solution. The most general real solution is the superposition of the two modes,

$$\theta(t) = A \sin \omega_1 t + B \cos \omega_1 t + C \sin \omega_2 t + D \cos \omega_2 t. \quad (18)$$

Solving (4) for z gives $z = -(2I/\varepsilon)\ddot{\theta} - (2\delta/\varepsilon)\theta$, and using $\delta/I = \omega_0^2$ together with $\ddot{\theta} = -\omega_1^2(\dots) - \omega_2^2(\dots)$ gives

$$z(t) = \frac{2I}{\varepsilon} \left[(\omega_1^2 - \omega_0^2)(A \sin \omega_1 t + B \cos \omega_1 t) + (\omega_2^2 - \omega_0^2)(C \sin \omega_2 t + D \cos \omega_2 t) \right]. \quad (19)$$

Initial conditions. For release from rest, $\dot{\theta}(0) = 0$ and $\dot{z}(0) = 0$ give

$$\omega_1 A + \omega_2 C = 0, \quad (\omega_1^2 - \omega_0^2)\omega_1 A + (\omega_2^2 - \omega_0^2)\omega_2 C = 0. \quad (20)$$

Eliminating $A = -(\omega_2/\omega_1)C$ yields $\omega_2(\omega_2^2 - \omega_1^2)C = 0$; since $\omega_2^2 \neq \omega_1^2$, necessarily $C = A = 0$: there are no sine terms. The remaining conditions $\theta(0) = \theta_0$ and $z(0) = z_0$ leave a 2×2 system for B and D ,

$$B + D = \theta_0, \quad (\omega_1^2 - \omega_0^2)B + (\omega_2^2 - \omega_0^2)D = \frac{\varepsilon z_0}{2I}, \quad (21)$$

whose solution is exactly (11). Substituting $A = C = 0$ into (18)–(19) recovers the closed-form equations of motion (9)–(10). ■

B Additional configurations

The package includes 14 predefined configurations spanning strong to weak coupling, different masses and inertias, and the periodicity condition. All satisfy the resonance condition (6) and can be reproduced with `./wilberforce <n>`. For the reference system, Figure 11 illustrates the bound on the integrator's energy drift.

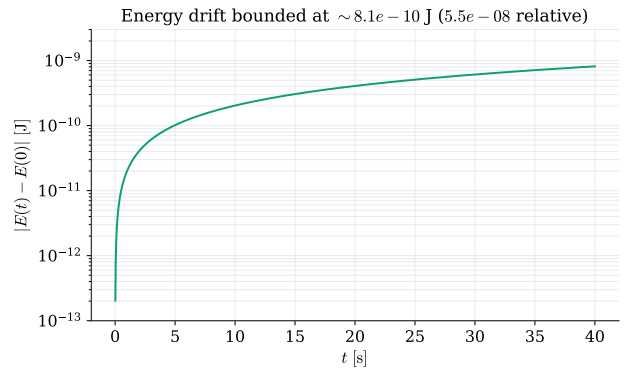


Figure 11: Absolute drift of the Fortran core's total energy throughout the integration, bounded at $\sim 8 \times 10^{-10}$ J ($\sim 5.5 \times 10^{-8}$ relative).

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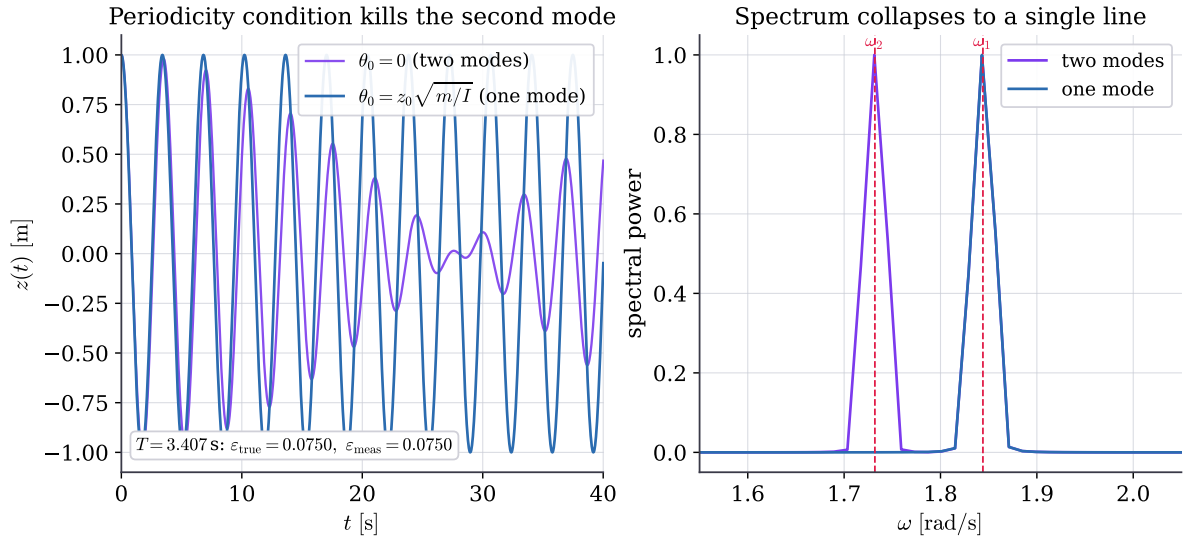


Figure 9: Measurement of ε . (Left) with $\theta_0 = 0$ the motion is not periodic; with $\theta_0 = z_0 \sqrt{m/I}$ it becomes single-frequency. (Right) the spectrum collapses from two peaks to one. A single period measurement recovers ε through eq. (15).

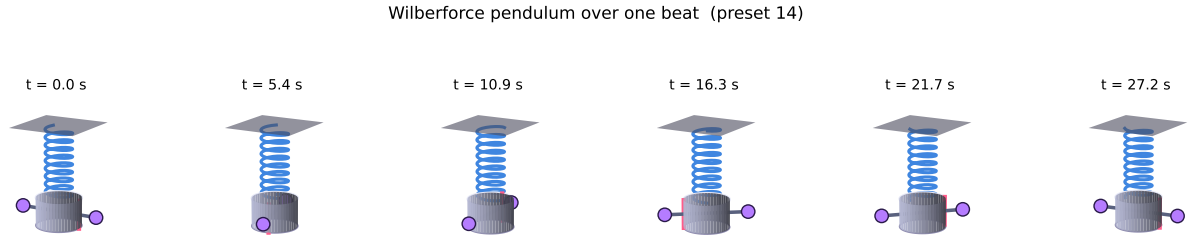


Figure 10: Sequence of the pendulum over a beat, rendered by the three-dimensional simulator. The rod's orientation and the stripe reveal rotation θ ; the mass's vertical position and the spring's extension reveal translation z .

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